

Quotient calculus for power–log leading terms

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2026-09-07

Abstract

Astra #14 rank 1. If $Z_\varphi(N) \sim C_\varphi N^{-\alpha_\varphi} (\log N)^{r_\varphi}$ and $Z_1(N) \sim C_1 N^{-\alpha_1} (\log N)^{r_1}$, then $Z_\varphi/Z_1 \sim (C_\varphi/C_1) N^{-(\alpha_\varphi-\alpha_1)} (\log N)^{r_\varphi}/(\log N)^{r_1}$ (`quotient_isEquivalent`), with the three consequences used for posterior expectations: equal leading terms give the constant limit C_φ/C_1 ; one fewer logarithm in the numerator gives decay like $1/\log N$, i.e. $(Z_\varphi/Z_1) \log N \rightarrow C_\varphi/C_1$; a strictly larger power in the numerator gives decay to 0. The calculus is independent of the chart indexing and uses totalised division. Zero `sorry/axiom`.

1 The things formalised

```
def powLog (C : ℝ) (r : ℝ) (N : ℝ) : ℝ := C * N ^ (-) * Real.log N ^ r
theorem quotient_isEquivalent (Z Z') (C C') (r r') (h : Z ~[atTop] powLog C r)
  (h' : Z' ~[atTop] powLog C' r') :
  (fun N => Z N / Z' N) ~[atTop]
  (fun N => (C / C') * N ^ (-(r - r')) * (Real.log N ^ r / Real.log N ^ r'))
theorem tendsto_quotient_of_eq ... : Tendsto (fun N => Z N / Z' N) atTop (C / C')
theorem tendsto_quotient_mul_log_of_eq ... (h : Z ~ powLog C (r + 1)) :
  Tendsto (fun N => Z N / Z' N * Real.log N) atTop (C / C')
theorem tendsto_quotient_zero_of_lt ... (h : C < 0) :
  Tendsto (fun N => Z N / Z' N) atTop (0)
```

2 Mathematics

`IsEquivalent.div` handles the quotient; the model term is simplified for $N > 1$ where $N^{-\alpha} > 0$ and $\log N > 0$. The zero limit uses $(\log N)^{r_\varphi} = o(N^\delta)$ for $\delta = \alpha_\varphi - \alpha_1 > 0$ (`Mathlib.isLittleO_log_rpow_rpow_atTop`) after bounding $(\log N)^{r_\varphi}/(\log N)^{r_1} \leq (\log N)^{r_\varphi}$ for $N \geq e$.

3 Paper-facing meaning

This is the deterministic backbone of `cor:empirical_expectation`'s leading term when numerator and denominator have different leading terms. Astra #14's correction is built in: an observable vanishing at the corner typically removes a logarithm rather than raising the power, so the posterior expectation decays like $1/\log N$ (`tendsto_quotient_mul_log_of_eq`), not like a power of N . The chart instantiation (leading-term extraction from the Taylor tree for equal starts, with $A_p \neq 0$ or $A_p = 0 \neq B_p$) is the next unit.

4 Lean notes

- `IsEquivalent.div` produces the Π quotient; convert with `congr_left` and a pointwise `rfl`. `IsEquivalent.tendsto_const` needs the model rewritten to `const` by `congr_right`.
- Avoid a global `div_eq_mul_inv/abs_of_pos _` rewrite: it hits the coefficient quotient C_φ/C_1 first; give `abs_of_pos` its argument explicitly.
- `squeeze_zero_norm'` with an eventual bound and `IsLittle0.tendsto_div_nhds_zero`.